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Optimizing calibration sample selection in infrared spectroscopy

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ABSTRACT

Chemometrics associated with infrared spectroscopy has been widely used in chemical analysis to establish predictive models. However, one of the challenges faced by the user when building and maintaining models is to select calibration samples. Multiple selection algorithms exist, but many rely on reference values or make assumptions about the underlying model. Among the available algorithms which rely solely on spectral data, Kennard-Stone is widely used in IR spectroscopy. Another distance-based selection algorithm, the Wootton-Sergent-Phan-Tan-Luu algorithm, emphasizes achieving uniform selection. However, both promising algorithms were originally designed to operate on datasets with low collinearity, and minimal overlapping features. These conditions are not met for IR spectral data. As a result, this raises the question of the relevance of applying them directly to raw spectra. To address this question, transformations such as spectral preprocessing, variable reduction, and encoding of variables are applied to mitigate spectral data complexity. This study evaluates how these three parameters affect model performance when samples are selected by KS and WSP from a candidate set. Results show dimensionality reduction and encoding generally provide slight improvements or maintain performance by reducing redundancy and promoting uniform spectral coverage. Preprocessing impacts algorithms differently: WSP is robust across strategies, whereas KS is more sensitive to preprocessing. Based on these findings, we recommend dimensionality reduction followed by encoding linearly structured data, and WSP for applications with uncertain preprocessing choices due to its greater stability. This work highlights the importance of data transformations before sample selection.

1. Introduction

Infrared (IR) spectroscopy is a widely used technique across many industries and scientific fields [1,2], including pharmaceuticals [3,4], agriculture and food processing [5,6], and recycling [7]. This technique stands out due to its ability to provide detailed information rapidly and non-destructively about spectral fingerprints, which are linked to properties of interest such as concentration, purity, or composition. However, to be effectively used in routine, IR spectroscopy must be coupled with chemometric models [1,8,9] that can interpret spectral information and provide predictions for the properties of interest. Calibration and model updating rely on datasets that pair IR spectra with reference values of the properties of interest, which are obtained through laboratory analyses. The acquisition of reference data often requires substantial effort and high costs due to the need for expert knowledge, expensive techniques, and labor-intensive procedures [10]. As a result, the construction of calibration datasets can represent a significant challenge for users and may limit the feasibility of certain applications. By contrast, spectral measurements are easy and inexpensive to obtain, which enable the collection of large numbers of spectra without major constraints. However, in many current practices, all collected spectra are systematically subjected to reference analyses, irrespective of their contribution to model development. This approach may result in the acquisition of information that is redundant, and thus an inefficient use of resources. When the question of sample selection is explicitly considered, most chemometric applications treat it as a preliminary step, where a fixed proportion of the available data is chosen a priori and assumed to be sufficient. In contrast, the present work adopts a different perspective. Rather than specifying a fixed number of calibration samples to ensure satisfactory model performance, we focus on

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identifying the minimal number of samples required to achieve a target predictive accuracy. To approach this minimum as closely as possible, we integrate sample selection directly into the training process. This perspective naturally leads to the question of which sample selection strategies are most effective.

Various sample selection algorithms exist and can broadly be divided into supervised and unsupervised approaches. Supervised methods rely on reference values to select samples. For example, the Simple Interval Calculation [11] (SIC) uses the interval prediction of reference values to assess influential samples. Similarly, SPY [12] (Sample set Partitioning based on Y) and SPXY [13] (based on both X and Y), as well as their variants, use reference values in the selection process. As a result, these approaches are suitable for choosing a subset of spectra from a pool of candidates that already have reference measurements. However, they do not address the problem of building a new model while minimizing the number of required reference analyses. In contrast, unsupervised methods rely solely on X to provide a representative coverage of the spectral space. Within this group, D-optimal design [14] is a classical approach that requires only X. Nevertheless, it is based on assumptions about the underlying model, which may not hold in the context of a new application. Other unsupervised methods are purely spectra-driven, including the Kennard-Stone [15] (KS) algorithm, widely recognized for its simplicity and effectiveness, the Wootton-Sergent-Phan-Tan-Luu [16] (WSP) algorithm, which emphasizes uniform coverage, and Puchwein's algorithm [17], which employs Mahalanobis distances to account for correlations among spectral variables. Clustering strategies [18] further extend this idea by grouping spectra and selecting representative samples from each cluster. Together, these unsupervised approaches provide practical solutions for reducing costly reference analyses, although their direct application to raw spectra is rarely questioned, despite the specific challenges posed by IR data.

While unsupervised approaches offer the advantage of relying solely on spectral variables, their performance depends strongly on the characteristics of the data. By their nature, spectra are characterized by highly redundant, correlated, and heterogeneous variables, with specific regions of interest. This can lead certain spectral regions to occupy a major role in the selection process. Therefore, unsupervised selection algorithms may preferentially emphasize regions with stronger signals or higher variance, which could be detrimental to calibrate the model. Furthermore, it has been suggested in a study [19] that a spectrum is influenced not only by the spectral signatures of the target compounds but also by external factors such as chemical interferences from other substances and physical phenomena like temperature fluctuations or light scattering. Consequently, a spectrum can be conceptually decomposed into two parts: a useful component, containing predictive information from the target analytes, and a detrimental component, consisting of noise, interferences, or irrelevant variability. Distinguishing between these components is crucial for enhancing the quality of the selection process and the reliability of the resulting calibration models.

Various transformations are commonly applied to mitigate detrimental variation and to facilitate model calibration. Techniques such as dimensionality reduction or spectral pretreatments are widely used to remove confounding effects and help models capture the relationships between spectra and the properties of interest. Several studies have investigated the role of such transformations in the context of sample selection. For instance, a paper [10] examined the effect of four parameters, including dimensionality reduction via principal component analysis (PCA) [20], selection algorithm, sample size and model complexity (based on Vapnik's theoretical framework). The study showed that, for a fixed sample size, input dimensionality and model complexity reduced the influence of the selection algorithm on model performance. However, model complexity was defined before training which in practice is challenging and may limit the model's ability to avoid underfitting or overfitting. Another study [21] focused on the effect of pretreatments coupled with Kennard Stone algorithm. The results showed that pretreatment had varying effects on the selection, which in turn influenced the model performance. While this work focused solely on the Kennard-Stone algorithm, it raises the question of whether other unsupervised selection algorithms would respond similarly to preprocessing strategies.

Finally, this paper is constructed as follows: the first part aims to systematically evaluate how PCA and encoding affect the performance of two widely used selection algorithms: KS and WSP. This is investigated using both scenarios. The second part of the paper aims to evaluate how pretreatment, PCA and encoding affect the performance of both algorithms on real-world cases involving milk and manure datasets. As final objective, this paper seeks to provide practical insights into how data transformations influence the ability of selection algorithms to identify informative calibration samples.

2. Materials and methods

2.1. Parameters influencing samples selection

In this study, we investigated the impact of three parameters on selection: variable reduction using PCA, preprocessing, and encoding. Each combination of parameters produced a specific configuration. For the simulated dataset, configurations followed the pattern: **Variable Reduction – Encoding**. For the real dataset, configurations followed the pattern: **Preprocessing – Variable Reduction – Encoding**.

The preprocessing parameter involved assessing the effect of two preprocessing strategies, Savitzky-Golay [22] (savgol) filtering, and Standard Normal Variate (SNV) [23] preprocessing. The parameters for Savitzky-Golay were set to apply a window length of 11 and a first-order derivative.

Variable reduction was performed using PCA [20], with the number of components determined based on a target percentage of variability explained rather than being set arbitrarily. A threshold of 99% variability was selected.

For the encoding parameter, each variable was scaled by subtracting its minimum value and then dividing by the range. This ensured that each variable was adjusted to values between 0 and 1. Encoding was applied either to raw spectra or PCA scores depending on the configuration.

All possible configurations are displayed in Table 1.

2.2. Selection algorithms

Selection algorithms were applied to the candidate sets to form multiple calibration sets to train a PLS model. Three selection algorithms were tested in this study, WSP, KS and random selection.

The WSP [16] algorithm used in this study begins by selecting the most central point from the available candidates. Then, it iteratively removes all samples that are within a predefined distance d. From the remaining candidates, it selects the nearest point and repeats the process until no more candidates are left. As a result, this algorithm does not directly control the number of selected samples since it depends on the distance d. The WSP algorithm was run multiple times to select a fixed number of samples for each calibration set. If multiple sets of the same size were possible, the one with the largest distance d was chosen.

The KS [15] algorithm used in this study starts by selecting the two points that are the farthest apart among all available samples. Then, it iteratively selects the sample that is the farthest from the already selected samples. For a set of k selected samples, the distance of each remaining sample is the minimum distance from the sample to each selected sample. The calibration set is obtained after selecting the desired n samples.

The random selection algorithm was included as an absence of selection strategy. It consists of randomly selecting n samples from the candidate set.

Table 1
Overview of preprocessing, dimensionality reduction, and encoding configurations.

Config. ID	Preprocessing	PCA	Encoding	Datasets
C1	None	No	No	Scenario 1
C2	None	Yes	No	Scenario 1
C3	None	Yes	Yes	Scenario 1
C4	None	No	Yes	Scenario 1
C5	None	No	No	Scenario 2
C6	None	Yes	No	Scenario 2
C7	None	Yes	Yes	Scenario 2
C8	None	No	Yes	Scenario 2
C9	None	No	No	Milk
C10	None	Yes	No	Milk
C11	None	Yes	Yes	Milk
C12	None	No	Yes	Milk
C13	None	No	No	Manure
C14	None	Yes	No	Manure
C15	None	Yes	Yes	Manure
C16	None	No	Yes	Manure
C17	Savitzky-Golay	No	No	Milk
C18	Savitzky-Golay	Yes	No	Milk
C19	Savitzky-Golay	Yes	Yes	Milk
C20	Savitzky-Golay	No	Yes	Milk
C21	Savitzky-Golay	No	No	Manure
C22	Savitzky-Golay	Yes	No	Manure
C23	Savitzky-Golay	Yes	Yes	Manure
C24	Savitzky-Golay	No	Yes	Manure
C25	SNV	No	No	Milk
C26	SNV	Yes	No	Milk
C27	SNV	Yes	Yes	Milk
C28	SNV	No	Yes	Milk
C29	SNV	No	No	Manure
C30	SNV	Yes	No	Manure
C31	SNV	Yes	Yes	Manure
C32	SNV	No	Yes	Manure

2.3. Evaluation strategy

For simulated datasets, PLS models were trained for a range of selected spectra comprised between 6 and 50 spectra increasing by 2 spectra for each configuration. This was repeated for the fifty simulations of each scenario, resulting in a total of 27 600 PLS models for all selection algorithms studied, as described in Table 2. Each set of selected spectra formed a dataset used to train a PLS model. To choose the optimal number of latent variables, multiple PLS models were tested on the validation set. The model with the best RMSE performance was selected as the final model for the tested number of spectra. The performances of models for each number of spectra tested were then compiled and compared to those of models trained with other selection algorithms, all using the same specific number of spectra. Finally, to assess whether differences in performance between configurations were statistically significant, a Mann-Whitney test was conducted. For each configuration tested, we obtained 50 RMSE values on the test set from 50 independent simulations. The Mann-Whitney test was then applied to compare the distributions of these 50 RMSE values between any two configurations. Results for the comparisons between PCA and RAW (Figs. 1–2), and the comparison between WSP and KS (Fig. 3) are presented in Table 4. Differences were considered statistically significant if the p-value was less than 0.05. For real datasets, the same strategy was employed, including the preprocessing parameter, over a range of 5 to 200 selected spectra, increasing by 2 spectra. It resulted in a total of 2880 PLS models trained for all selection algorithms studied, as described in Table 3. Preprocessing methods were applied solely during the selection process, while calibration models were trained on raw spectra in order to specifically assess their impact on selection. All selection strategies were carried out exclusively on the calibration set, the validation set was used only for model tuning, and the test set was reserved solely for final performance comparison.

Table 2
Summary of PLS Models Trained for simulated data.

Parameter	Values/Levels	Count
Number of selected spectra	From 6 to 50 (step of 2)	23
Variable reduction	With PCA, Without PCA	2
Encoding	With encoding, Without encoding	2
Selection algorithms	WSP, KS, Random	3
Simulation scenarios	Two scenarios	2
Repetitions	Simulations	50
Total models trained	$23 \times 2 \times 2 \times 3 \times 2 \times 50$	**27 600**

2.4. Data

2.4.1. Simulated dataset

Data were simulated using the framework proposed in Ref. [24], this simulation process is based on the concept of useful and detrimental information contained in a spectrum described in Ref. [19]. A total of n spectra, each with p variables, and their associated values were constructed as defined in Eq. (1) to Eq. (4). Eq. (2) and Eq. (3) result from the decomposition of Eq. (1). In this section, we adopt the following conventions: scalars are denoted by italic lowercase letters such as a; vectors are represented by bold lowercase letters such as $\mathbf{y}$; matrices are indicated by bold uppercase letters, such as $\mathbf{X}$; finally, functions are written in italic with parentheses, for instance $g()$.

$$\mathbf{X} = \mathbf{X_U} + \mathbf{X_D} + \mathbf{E} \quad \text{Eq. 1}$$

$\mathbf{X}_{(n,p)}$ represents the spectra, $\mathbf{X_U}_{(n,p)}$ the useful part of the spectra, $\mathbf{X_D}_{(n,p)}$ the detrimental one and $\mathbf{E}_{(n,p)}$ the noise (e.g., Gaussian noise).

$$\mathbf{X_U} = \mathbf{T_U} \cdot \mathbf{P'_U} \quad \text{Eq. 2}$$

$$\mathbf{X_D} = \mathbf{T_D} \cdot \mathbf{P_D}' \quad \text{Eq. 3}$$

Where $\mathbf{T_U}_{(n,u)}$ are the scores of the loadings $\mathbf{P'_U}_{(u,p)}$ for the useful part, and $\mathbf{T_D}_{(n,d)}$ are the scores of the loadings $\mathbf{P'_D}_{(d,p)}$ for the detrimental part. u and d denote the number of useful and detrimental structures.

$$\mathbf{y} = g(\mathbf{T_U}) + \mathbf{f} \quad \text{Eq. 4}$$

$\mathbf{y}_{(n,1)}$ represents the concentration values, g is the function that links the contributions of the column structures from $\mathbf{X_U}$ to $\mathbf{y}$, $\mathbf{f}_{(n,1)}$ is the noise for the reference measure (e.g: Gaussian).

$$\mathbf{X} = \mathbf{T_U} \cdot \mathbf{P'_U} + \mathbf{T_D} \cdot \mathbf{P'_D} + \mathbf{E} \quad \text{Eq. 5}$$

Sets of spectra were simulated as follows.

- Pure spectra ($\mathbf{P_U}$) were obtained with classical least squares [24] from spectra of a mixture of glucose/ethanol, $\mathbf{P_D}$ are combinations of random Gaussian curves representing artificial chemicals
- Score matrices ($\mathbf{T_U}$ and $\mathbf{T_D}$) were built using a random Gaussian distribution with a mean of 20 and a standard deviation of 5 for $\mathbf{T_U}$ and using a random Gaussian distribution with a mean of 0 and a standard deviation of 1 for $\mathbf{T_D}$
- Noise matrix $\mathbf{E}$, representing instrumental noise was obtained by following a normal multivariate distribution $N_{(0,\ sigma)}$, with sigma = 10% of the total signal range.
- $\mathbf{X}$ was built with Eq. 5
- $\mathbf{y}$ was built from $\mathbf{T_U}$ according to the following relation: $\mathbf{y} = k \cdot \mathbf{T_U} + \mathbf{f}$, with $\mathbf{f}$ following a normal distribution $N_{(0,sigma)}$, with sigma = 10% of the standard deviation of $\mathbf{y}$, and $k = 10$.

Fifty sets of 500 spectra were simulated using this framework, each simulation was a mixture involving two pure spectra: ethanol and glucose. Spectra were simulated from 1100 nm to 2500 nm with a spectral resolution of 2 nm and were associated with concentration values, which served as responses of interest. Multiple simulations were

performed instead of one to ensure that the result was not biased due to a favorable simulation or unfavorable simulation due to the random part aspect of it. Across the 50 simulations, all the Y values were kept identical to allow the compilation of the results into means for enhancing robustness. Finally, Y was composed of glucose and ethanol concentrations. The simulation process was designed so that $\mathbf{X_U}$ of glucose was intentionally set to be half that of $\mathbf{X_U}$ of ethanol, creating two distinct scenarios: scenario 1, where $\mathbf{X_U}$ is in major proportion, and scenario 2 where $\mathbf{X_U}$ is in minor proportion. Scenario 1 represents an optimal case for selecting spectra based on Euclidean distances, where the useful part is in major proportion. In contrast, Scenario 2 represents a more challenging case, where the useful part is in minor proportion, making it more difficult to select spectra effectively. Finally, the dataset was randomly split into a candidate set and an intermediary set, with respectively 400 and 100 spectra.

2.4.2. Real dataset: manure and milk datasets

Two datasets from the agrofood industry were chosen to confirm the findings and trends established with simulated data and assess the influence of the preprocessing parameter.

The first dataset consisted of milk samples, and contained 395 NIR spectra ranging between 900 nm and 1700 nm with a resolution of 3 nm [25]. The acquisition of the spectra was done in 2 periods, resulting in 2 datasets. As a result, the first period comprised 316 spectra which constituted the candidate set while the second period comprised 79 spectra, constituting an intermediary set. The intermediary set was split randomly into a validation set and a test set (39-40 respectively). The milk originated from Holstein cows, and each sample was collected using an automated on-farm milking system equipped with a near-infrared (NIR) sensor operating in transmittance mode. The sensor system recorded 100 measurements per sample, which were averaged to improve the signal-to-noise ratio. During the acquisition, spectral signals were automatically linked to reference analyses, ensuring that each milk spectrum corresponded to its respective chemical composition without manual handling. The time frame between the acquisition periods was two weeks. For this study, the chemical constituent of interest was lactose, which has been shown to be more difficult to predict than fat or protein due to its weaker interaction with NIR radiation and the dominant absorbance of water and fat in milk.

The second dataset consisted of pig manure samples, and contained 584 NIR spectra ranging between 426 and 1686 nm with a resolution of 9 nm [26]. These samples had been stored for over a year under the same conditions. Each sample was scanned in reflectance mode using a Zeiss Corona 45 vis-NIR fibre diode array instrument with a remote measuring head, which allowed measurements through a window close to the manure without exposing the sensor to the sample. Approximately 60 mL of manure was placed in a 50 mm diameter, 3 cm high homemade cup, stirred manually before measurement, and re-stirred between successive measurements to avoid settling. Three measurements of two scans each (1 s per scan) were taken per sample to obtain a representative spectrum. A calibration set of 420 samples was collected in the spring of 2004 from multiple farms in Flanders, Belgium. These samples were stored in airtight containers at 4 °C from the time of sampling until analysis and were thoroughly stirred before measurement to ensure homogeneity. The remaining 164 samples, collected in spring 2003, were split evenly into a test set and a validation set (82 samples each). In this study, the chemical constituent of interest was dry matter content, which is known to be the dominant fingerprint influencing the spectral signature.

3. Results

This results section is organized into two parts. The first part presents the results obtained on the simulated datasets with Figs. 1–3. The second part focuses on the results for the Milk and Manure datasets with Figs. 4–9.

3.1. Simulated data

Fig. 1a to d illustrate the mean RMSE across all simulations for each test dataset, under the configurations **C1** to **C8** applied to the WSP selection algorithm, over a range of 6 to 50 selected spectra. For all the following figures, the blue curve with squares represents the RMSE evolution for the raw configuration, while the red curve with dots corresponds to the PCA configuration.

In Fig. 1a, RMSE values range from 0.471 to 0.114 for the blue curve with squares representing the configuration without PCA and from 0.526 to 0.114 for the red curve with dots corresponding to the configuration with PCA. A significant drop in RMSE is observed between 6 and 20 selected spectra, with a plateau occurring at around 30 selected spectra. Concerning the comparison between the red curve with dots and the blue curve with squares, five values for 10, 16, 24, 28, and 40 selected spectra show a significant difference in favor of the red curve with dots, all associated with a $r^2 > 0.9$. One value shows a significant difference which was not in favor of the red curve with dots, observed at 6 spectra and associated with a $r^2 < 0.8$.

From these observations, the encoding of PCA scores provides better performance than the direct encoding of raw spectra. One hypothesis that explains this difference is that the scores reduce the impact of physical effects by distributing weight more evenly across the components. Consequently, encoding brings the selection closer to a Mahalanobis-like distance, which accounts for the variance structure of the data and enables a more uniform exploration, in line with the principles behind WSP. This improvement can be attributed to the fact that the encoding is applied to orthogonal data. The PCA components provide uncorrelated and independent directions, allowing for a more balanced representation of information across components. In contrast, raw spectra are collinear, making encoding less effective and less meaningful, as it does not account for redundancy among spectral variables or the multiple constituents of the sample.

A similar pattern is observed in Fig. 1c, with differences appearing even more clearly due to the more challenging conditions of Scenario 2. In Fig. 1c, RMSE values span from 0.743 to 0.127 for the blue curve with squares and from 0.788 to 0.124 for the red one. A notable RMSE decrease is observed up to 20 selected spectra, reaching a plateau around 40. Seven selections (8, 10, 12, 14, 28, 44, 50) show significant improvement with PCA, all associated with a $r^2 > 0.9$. This effect becomes clearer in Scenario 2, which is designed to include spectra with more detrimental compounds than those in Scenario 1. The useful compound corresponds to the spectrum of interest, while the detrimental compounds include all other contributions, such as additional spectra, external interferences, or noise. These detrimental components can obscure the signal of interest and affect the efficiency of the selection. Therefore, this configuration allows the selection algorithm to explore the spectral space more evenly, reducing the risk that a single spectrum disproportionately drives the outcome, thereby amplifying the effect of encoding PCA scores.

In Fig. 1b, RMSE values vary between 0.429 and 0.113 for the blue curve with squares and from 0.429 to 0.114 for the red curve with dots. The global behavior is similar to Fig. 1a, with a significant drop until 20 spectra are selected and a plateau reached around 30. No statistically significant differences are found between configurations, indicating similar performance between the two configurations in Scenario 1.

This can be explained by the fact that when PCA scores are not encoded, the space defined by the principal components does not allow for uniform exploration, as each component has a different amplitude. By construction, the first principal component exhibits the largest amplitude, dominating the selection-based distances. As a result, spectra represented by PCA scores or by raw variables occupy essentially the same positions relative to other spectra. Thus, PCA without encoding provides limited improvement compared to raw spectra.

A similar pattern is observed in Fig. 1d, with differences appearing slightly more pronounced due to the more challenging conditions of

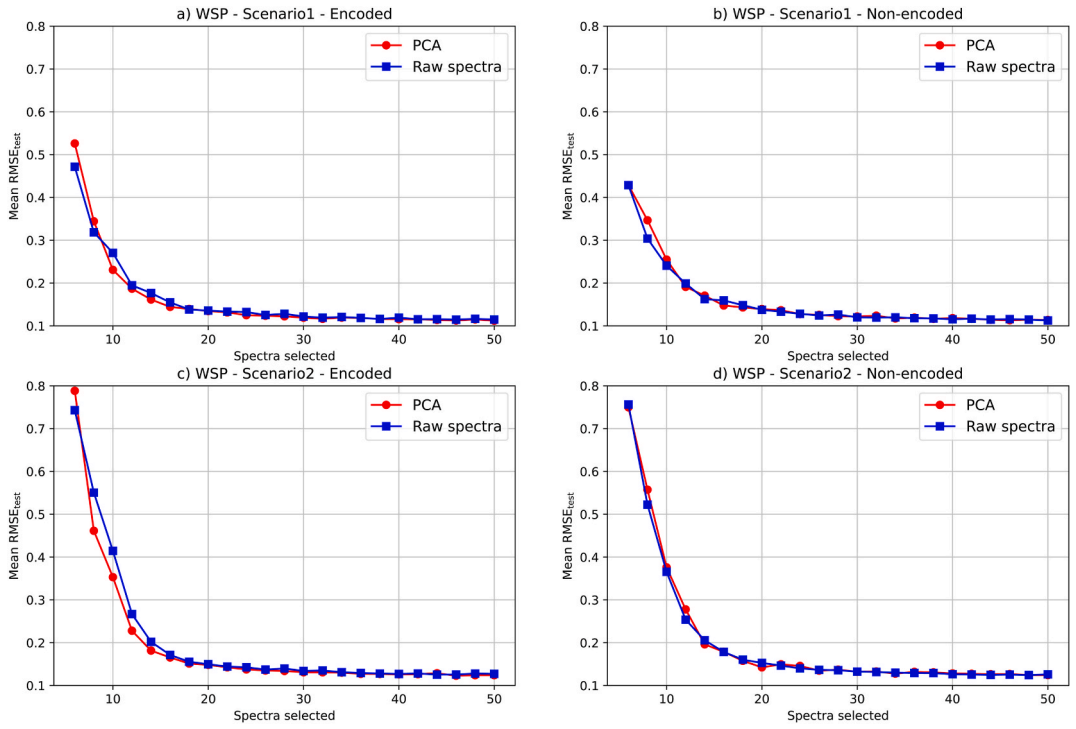


Fig. 1. Plot of the mean RMSE computed across all simulations on each test datasets for various configurations using WSP **a)** Configurations C3 and C4; **b)** Configurations C1 and C2; **c)** Configurations C7 and C8; **d)** Configurations C5 and C6. Red curves with dots correspond to the configurations with PCA, while blue curves with squares correspond to the configurations without PCA. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

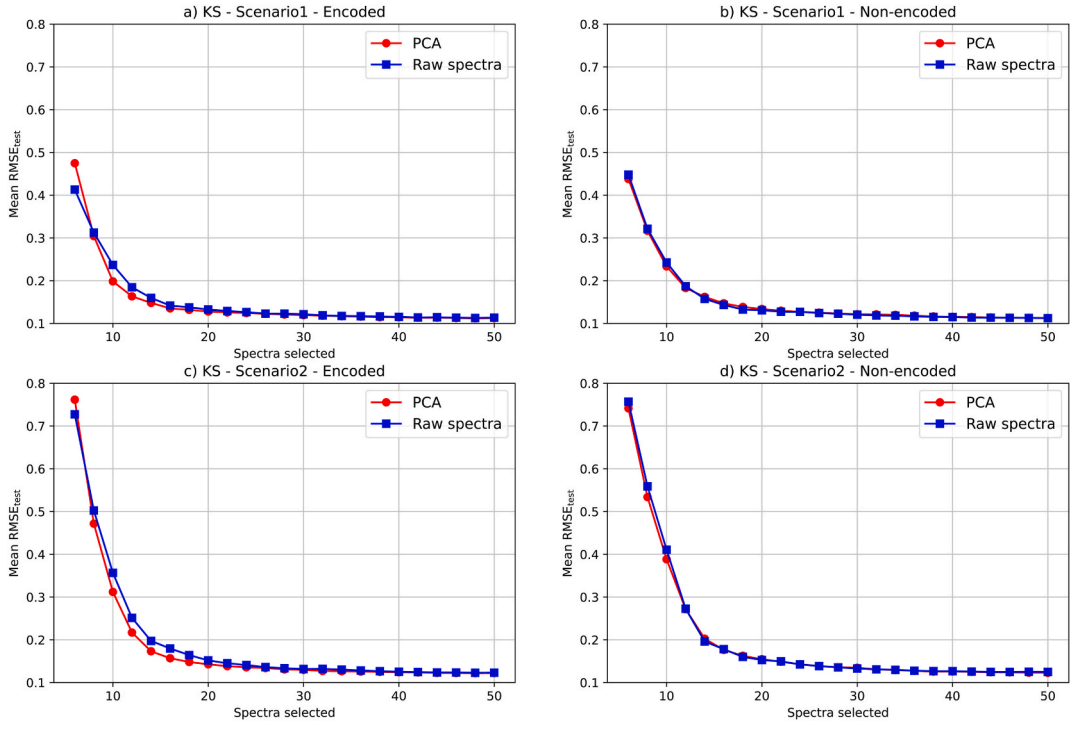


Fig. 2. Plot of the mean RMSE computed across all simulations on each test datasets for various configurations using KS **a)** Configurations C3 and C4; **b)** Configurations C1 and C2; **c)** Configurations C7 and C8; **d)** Configurations C5 and C6. Red curves with dots correspond to the configurations with PCA, while blue curves with squares correspond to the configurations without PCA. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

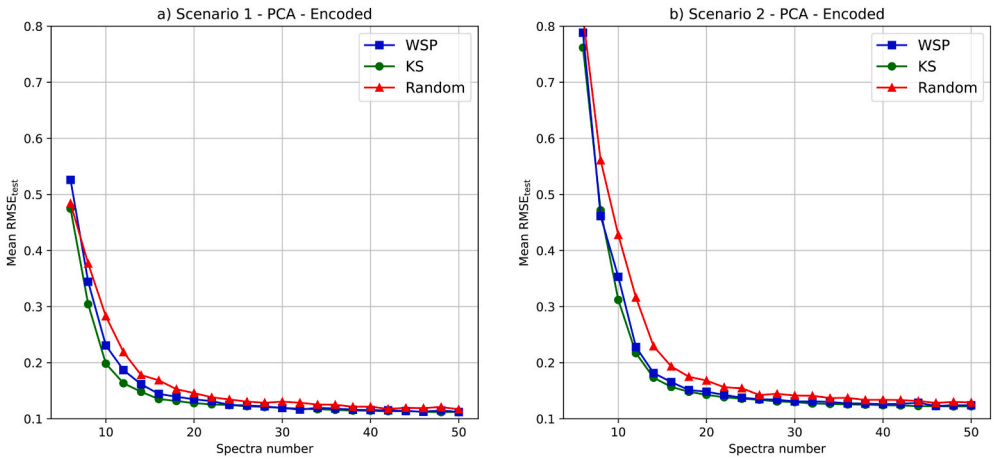


Fig. 3. Plot of the mean RMSE computed across all simulations on each test datasets for the best configurations using **KS, WSP** and **random** selection algorithms; **a)** C3 configurations for both algorithms; **b)** C7 configuration for both algorithms. Red curves with dots correspond to performance observed with random selection, while the blue ones correspond to the selection made with **WSP** and finally the green ones to the **KS** selections. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 3
Summary of PLS Models Trained for Manure and Milk datasets.

Parameter	Values/Levels	Count
Number of selected spectra	From 5 to 200 (step of 5)	40
Variable reduction	With PCA, Without PCA	2
Encoding	With encoding, Without encoding	2
Selection algorithms	WSP, KS, Random	3
Preprocessing	Savgol, no pretreatment, SNV	3
Datasets	Milk, Manure	2
Total models trained	$40 \times 2 \times 2 \times 3 \times 3 \times 2$	**2880**

Scenario 2. RMSE ranges from 0.756 to 0.126 for the blue curve with squares and 0.750 to 0.124 for the red one, with performance stabilizing around 40 selected spectra. Two values with significant differences are noted for 12 and 20 spectra ($r^2 > 0.9$). One value shows an improvement with the use of PCA (20), while the other shows a degradation in performance (12). Again, the absence of encoding limits the benefit of PCA, but the effect is accentuated in this scenario due to the increased difficulty and the less dominant component of interest.

Considering this interpretation, comparing Fig. 1a and b highlights the impact of encoding on the performance of PCA scores versus raw spectra. In Fig. 1a, encoding the PCA scores improves performance, whereas in Fig. 1b, when the encoding effect is removed, no significant differences are observed. This indicates that the improvements seen in Fig. 1a are not due to PCA alone, but to the combined effect of PCA and encoding. The same reasoning applies to the comparison between Fig. 1c and d. However, the differences are more pronounced due to the increased difficulty of Scenario 2, which amplifies the impact of encoding PCA scores on the selection process. In both scenarios, PCA without encoding provides comparable performance to raw spectra.

Taken together, these results confirm that the optimal WSP configuration for this studied case combines PCA with encoded scores. Encoding balances the contributions of all PCA components, reducing the dominance of the first components and mitigating the impact of physical effects. This brings the selection closer to a Mahalanobis-like distance and allows a more uniform exploration consistent with the principles of WSP. Importantly, the effect is more pronounced in Scenario 2 than in Scenario 1, due to the less dominant component of interest, which amplifies performance differences. Statistical analysis confirms that encoding offered a beneficial or neutral impact in all configurations, particularly when coupled with PCA and more pronounced in Scenario 2.

The same analysis is conducted on KS and is shown in Fig. 2a–d.

Fig. 2a to d illustrate the mean RMSE across all simulations for each test dataset, under the configurations **C1** to **C8** applied to the KS selection algorithm, over a range of 6 to 50 selected spectra.

In Fig. 2a, RMSE values range between 0.412 and 0.113 for the blue curve with squares and 0.474 to 0.112 for the red curve with dots. RMSE decreases significantly between 6 and 20 selected spectra, reaching a plateau around 30 selected spectra. Concerning the comparison between PCA and raw spectra, four selections (10, 12, 14, 18) show significant differences, all associated with $r^2 > 0.9$. One selection at 6 spectra shows a minor negative impact ($r^2 < 0.8$). Overall, this figure demonstrates the same observations noted in Fig. 1a, with slightly better performance for fewer than 20 selected spectra.

Consequently, applying PCA followed by encoding the PCA scores provides equal or better performance than encoding raw spectra in Scenario 1. As observed with WSP, the improvement arises from the combined effect of PCA and encoding, which balances the contributions of all components and reduces the influence of detrimental or redundant spectral variables. The similarity with WSP can be attributed to the shared reliance on Euclidean distances, producing comparable behaviors under similar configurations (Figs. 1a–2a).

Table 4
Number of spectra showing significant differences ($p < 0.05$) across 50 simulations for each comparison.

Comparison	Figure	Spectra where the **first** strategy significantly outperformed the **second**	Spectra where the **second** strategy significantly outperformed the **first**
PCA vs RAW	1a	10, 16, 24, 28, 40	6
PCA vs RAW	1b	–	–
PCA vs RAW	1c	8, 10, 12, 14, 28, 44, 50	–
PCA vs RAW	1d	20	12
PCA vs RAW	2a	10, 12, 14, 18	6
PCA vs RAW	2b	–	–
PCA vs RAW	2c	12, 14, 16, 18, 32, 34	–
PCA vs RAW	2d	–	–
WSP vs KS	3a	10, 12, 14, 16, 18, 20, 22, 48	–
WSP vs KS	3b	32, 44	–

In Fig. 2b, RMSE values range between 0.448 and 0.112 for the blue curve with squares and 0.438 to 0.113 for the red curve with dots. RMSE decreases significantly between 6 and 20 selected spectra, reaching a plateau around 30. No statistically significant differences are observed between PCA and raw spectra. As in Fig. 1b, this confirms that applying PCA without encoding provides limited improvement for KS in Scenario 1. When PCA scores are not encoded, the space defined by the principal components is not uniform: the first component dominates the selection distances, and spectra represented by PCA scores or raw variables occupy essentially the same relative positions. These observations indicate that PCA without encoding offers little advantage over raw spectra, while still providing some computational efficiency.

In Fig. 2c, RMSE values range between 0.727 and 0.123 for the blue curve with squares and 0.761 to 0.122 for the red curve with dots. RMSE decreases significantly between 6 and 20 selected spectra, reaching a plateau around 40. Six selections (12, 14, 16, 18, 32, 34) show significant improvements with PCA encoding ($r^2 > 0.9$). Overall, this figure demonstrates the same observations as Fig. 2a, with differences appearing slightly more pronounced due to the more challenging conditions of Scenario 2. By design, Scenario 2 contains spectra with a higher proportion of detrimental compounds compared to Scenario 1. These detrimental components can obscure the signal of interest, reducing the efficiency of the selection process. Consequently, encoding PCA scores amplifies the effect, enabling the selection algorithm to better capture relevant variations.

In Fig. 2d, RMSE values range between 0.757 and 0.125 for the blue curve with squares and 0.741 to 0.123 for the red curve with dots. RMSE decreases significantly between 6 and 20 selected spectra, reaching a plateau around 40. No statistically significant differences are observed between PCA and raw spectra. As with Fig. 2b, PCA without encoding provides comparable performance to raw spectra. However, the effect of the first component is more pronounced due to the increased difficulty of Scenario 2, meaning spectra represented by PCA scores or raw variables occupy essentially the same relative positions, consistent with KS behavior.

Comparing Fig. 2a and b highlights the impact of encoding on PCA performance. In Fig. 2a, encoding PCA scores improves performance, whereas in Fig. 2b, removing the encoding leads to no significant differences. Similarly, Fig. 2c and d follow the same pattern: encoding PCA scores improves or maintains performance, with effects amplified in Scenario 2 due to the higher difficulty and less dominant component of interest. These observations confirm that the performance gains observed in Fig. 2a and c are due to the combined effect of PCA and encoding, rather than PCA alone.

Across Fig. 1a–d and 2a-d, trends are consistent for both WSP and KS. The combination of PCA with encoding systematically improves or at least maintains performance compared to directly encoding raw spectra. The beneficial effect arises because encoding PCA scores allows a more uniform exploration of the PCA space, whereas raw spectra remain redundant and less influenced by encoding. Applying PCA without encoding produces no meaningful improvement, as the relative structure of the spectra is preserved. Differences between configurations are more pronounced in Scenario 2 than in Scenario 1, reflecting the increased difficulty where the spectrum of interest is less dominant. These observations confirm that the optimal KS configuration, as with WSP, combines PCA with encoded scores.

Building on these observations, Fig. 3 further focuses on the comparison of the two algorithms under their best-performing configurations.

Fig. 3a to b shows the mean RMSE across all simulations for each test dataset, with the **C3** and **C7** configurations for each selection algorithm over a range of 6 to 50 selected spectra. This figure allows comparison of KS and WSP on their best configuration tested with the random selection that represents no selection algorithm used.

Fig. 3a and b shows the RMSE evolution for WSP (blue squares), KS (green), and random selection (red dots). In both cases, RMSE decreases sharply between 6 and 20 selected spectra and reaches a plateau around 40. Both WSP and KS outperform random selection across most of the range, with statistically significant differences from 6 to 30 spectra in Fig. 3a and from 6 to 40 spectra in Fig. 3b. KS outperforms WSP for selections between 10 and 22 and at 48 spectra in Fig. 3a, and at 32 and 44 spectra in Fig. 3b, highlighting scenarios where KS provides an advantage over WSP.

Firstly, all algorithms reach a plateau in performance once enough spectra have been selected, as additional selections yield progressively less new information. The difference between random selection and both distance-based algorithms can be explained by the fact that randomly selected spectra do not guarantee to select similar spectra in terms of information gained. In contrast, distance-based selection algorithms ensure that the selected spectra are sufficiently spread out in the spectral space.

Finally, differences between KS and WSP in Fig. 3a can be attributed to the nature of the Kennard-Stone algorithm, which tends to select extreme samples by design. Such an approach is advantageous when most of the spectral information is relevant and when the relationship between spectral variables and the target compounds is predominantly linear. In this case study, PLS models were selected for their ability to capture linear relationships, which tends to favor KS. Therefore, the observed difference can be explained by the suitability of Kennard-Stone to this specific application. As a final point, between 22 and 30 selected spectra, both algorithms exhibit comparable behavior before the plateau. This can be explained by the fact that both algorithms rely on Euclidean distances, and that KS tends to achieve relatively uniform selections once enough spectra are chosen.

Compared to Fig. 3a, the algorithms show more similar behavior in scenario 2 (Fig. 3b). This can be explained by the selection strategy of KS, which focuses on extreme samples. When detrimental or irrelevant information is predominant, this strategy becomes less effective. In contrast, WSP does not rely on such assumptions and instead explores the variable space more uniformly, making it more robust in situations where informative features are not concentrated at the extremes. As a result, in this context, both algorithms tend to perform more similarly. In conclusion, on simulated data, WSP appears as a valuable alternative to the Kennard-Stone algorithm, exhibiting comparable performance. One of its main advantages lies in its assumption-free nature and its ability to uniformly explore the variable space, which proves particularly beneficial in cases involving non-linear relationships. Another key insight from this study is that variable encoding can enhance selection performance which is a result that may seem counterintuitive to chemometricians.

To extend this study, both selection algorithms are next applied to real-world datasets, combined with PCA-based encoding and preprocessing. Our goal is to evaluate their effectiveness in real applications and confirm our findings on simulated data.

3.2. Real data

3.2.1. Milk dataset

Fig. 4a–d shows RMSE of the test dataset for milk dataset, under the configurations C9 to C12, C17 to C20 and C25 to C28 using WSP across a range of 40 to 200 selected spectra.

For Fig. 4a, RMSE values for SNV preprocessing range from 0.098 to 0.056, while values for no pretreatment range from 0.094 to 0.060 and Savitzky-Golay preprocessing from 0.094 to 0.060. RMSE decreases notably between 40 and 100 selected spectra, reaching stable performance at 115 selected spectra. Similarly, Fig. 4b shows RMSE values ranging from 0.102 to 0.058 for SNV, from 0.092 to 0.060 for no pretreatment, and from 0.118 to 0.058 for Savitzky-Golay. The decrease in RMSE occurs between 40 and 100 selected spectra, with stabilization at 145 spectra. Across these two figures, no preprocessing method consistently outperforms the others. SNV shows slightly lower performance in some intermediate ranges, but overall, the effect of preprocessing varies

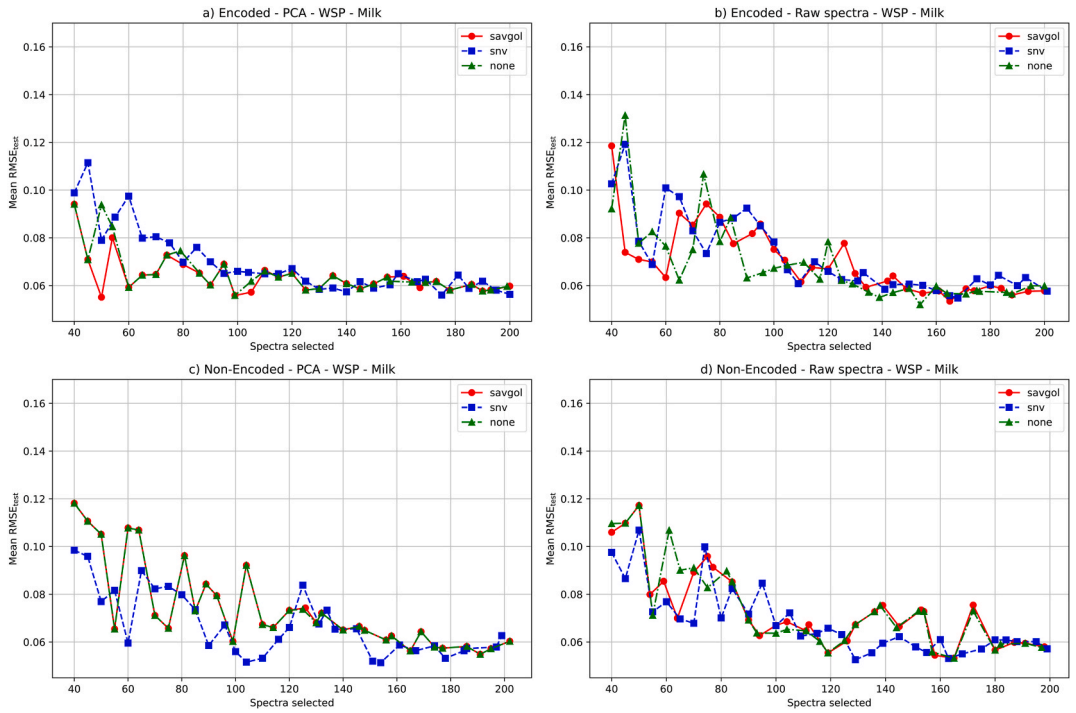


Fig. 4. Plot of the mean RMSE test on the milk dataset for various configurations using WSP **a)** Configurations C11, C19 and C27; **b)** Configurations C12, C20 and C28; **c)** Configurations C10, C18 and C26; **d)** Configurations C9, C17 and C25. Red curves with dots correspond to the configurations with savgol, while the blue curves with squares correspond to snv and the green ones to no pretreatment. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

depending on the configuration. Moving to Fig. 4c, RMSE ranges from 0.098 to 0.062 for SNV, while both no pretreatment and Savitzky-Golay range from 0.118 to 0.060. RMSE decreases between 40 and 100 selected spectra, stabilizing at 175 spectra. Unlike the previous figures, SNV reaches peak performance at several points, including 40, 45, 50, 80, and 100-120 selected spectra. Finally, Fig. 4d presents RMSE values from 0.097 to 0.057 for SNV, from 0.110 to 0.057 for no pretreatment, and from 0.108 to 0.058 for Savitzky-Golay. Stabilization occurs at 180

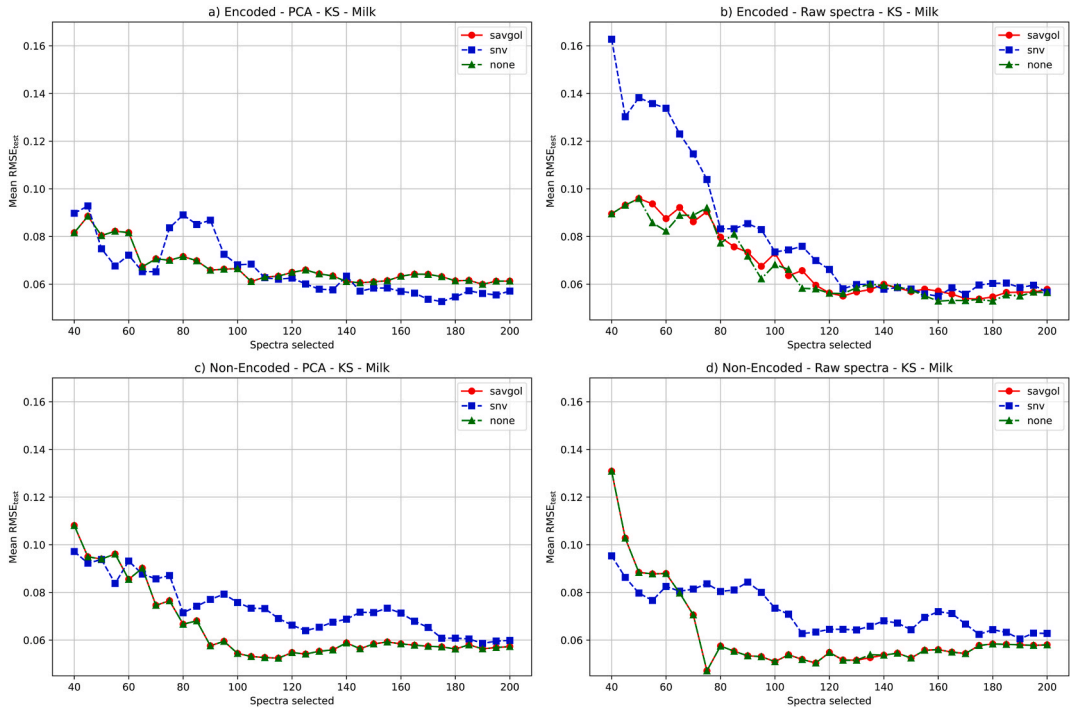


Fig. 5. Plot of the mean RMSE test on the milk dataset for various configurations using KS **a)** Configurations C11, C19 and C27; **b)** Configurations C12, C20 and C28; **c)** Configurations C10, C18 and C26; **d)** Configurations C9, C17 and C25. Red curves with dots correspond to the configurations with savgol, while the blue curves with squares correspond to snv and the green ones to no pretreatment. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

selected spectra, with SNV showing peak performance at 40-50 and 130-150 spectra. Overall, the three methods display similar behavior across the selection range.

Across all figures, the main factor affecting RMSE stabilization is the use of encoding and dimensionality reduction. Configurations with encoding stabilize earlier, between 110 and 145 selected spectra, compared with non-encoded configurations, which stabilize at 175-180 spectra. PCA-based configurations also converge faster than raw spectra, indicating that dimensionality reduction captures the most relevant variance and reduces redundancy. Preprocessing effects are variable. No single method consistently outperforms others; SNV occasionally shows peaks or reduced performance depending on the selected spectra range, while Savitzky-Golay and no pretreatment perform similarly. These observations indicate that preprocessing impacts are configuration-dependent and that the selection algorithm, alongside encoding, determines the ultimate effect. For practitioners, systematically applying PCA and encoding accelerates stabilization without compromising predictive accuracy.

The same analysis is conducted on KS and is shown in Fig. 5a–d.

Fig. 5a–d shows RMSE of the test dataset for milk dataset, under the configurations C9 to C12, C17 to C20 and C25 to C28 using KS across a range of 40 to 200 selected spectra.

For Fig. 5a, RMSE values for SNV preprocessing range from 0.090 to 0.057, while no pretreatment and Savitzky-Golay both range from 0.081 to 0.061. RMSE decreases notably between 40 and 100 selected spectra, with stabilization observed around 110 spectra. KS selection shows distinct behavior: no pretreatment and Savitzky-Golay yield identical selections due to PCA requiring only two components to explain 99 % of variance, whereas SNV outperforms after 145 spectra, indicating higher sensitivity of KS to preprocessing.

Fig. 5b shows RMSE values from 0.162 to 0.057 for SNV, and from 0.090 to 0.057 for no pretreatment and Savitzky-Golay. RMSE decreases between 40 and 100 spectra, stabilizing around 125-130 spectra. Here, SNV differs from the other two preprocessing methods for 40-80 spectra due to incompatibility with encoding parameters, while no pretreatment and Savitzky-Golay exhibit similar trends.

For Fig. 5c, RMSE ranges from 0.097 to 0.060 for SNV, while no pretreatment and Savitzky-Golay both range from 0.108 to 0.057. RMSE decreases between 40 and 100 spectra, with stabilization around 180 spectra. In this case, no pretreatment and Savitzky-Golay outperform SNV, likely because SNV removes relevant spectral information.

Fig. 5d shows RMSE values from 0.095 to 0.063 for SNV, and from 0.130 to 0.058 for no pretreatment and Savitzky-Golay. RMSE decreases between 40 and 100 spectra, reaching a plateau around 200 spectra. Peak performance for KS is observed at 75 spectra, guiding optimal comparison with WSP. SNV consistently lags, while no pretreatment and Savitzky-Golay perform better.

Across Fig. 5a–d, encoded configurations (**5a-b**) stabilize earlier than non-encoded ones (**5c-d**), at 110-130 versus 180-200 spectra, respectively. PCA-based selection accelerates stabilization compared with raw spectra, highlighting the advantage of dimensionality reduction. Preprocessing effects vary: SNV benefits from encoding but may reduce performance without it, whereas no pretreatment and Savitzky-Golay preserve relevant information. Overall, encoding and dimensionality reduction improve both predictive performance and convergence speed, though KS may achieve peak performance in non-encoded configurations, indicating algorithm-specific effects.

Fig. 6 shows the mean RMSE on the milk dataset, with C11 for WSP and C9 for KS over a range of 40 to 200 selected spectra. This figure allows comparison of KS (green curve) and WSP (blue curve) on their best configuration tested with the random selection (red curve). RMSE decreases significantly between 40 and 100 selected spectra until reaching a plateau for 180 selected spectra for all 3 algorithms. As a first observation, KS and WSP are globally better in performance than random selection in the range of number of selected spectra studied. Another key observation is that KS in this studied case performs slightly better than WSP on the 75 to 175 spectra range.

This behavior can be interpreted in terms of how the two algorithms explore the spectral space. KS tends to select extreme samples, which is advantageous for this predominantly linear dataset, whereas WSP distributes selections more evenly. Consequently, the relative performance of the two algorithms depends on the preprocessing applied. WSP is generally more robust to preprocessing variations, making it a reliable choice when the optimal preprocessing is unknown. As seen in simulated data, PCA and encoding further benefit WSP by reducing dimensionality and filtering noise, supporting its strategy of uniform coverage. In contrast, KS performs better on raw spectra, as PCA or encoding can distort the extremes it relies on, highlighting that the effectiveness of the selection strategy can be dataset- and algorithm-specific.

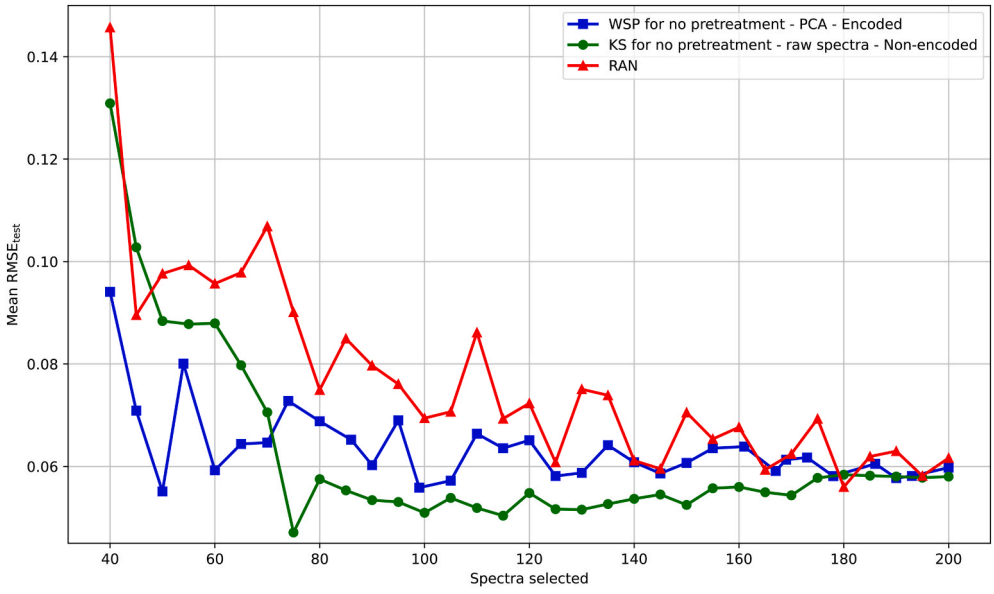


Fig. 6. Plot of RMSE test on the milk dataset for the best configurations using KS, WSP and random selection algorithms; C11 using WSP; b) C9 using KS. Red curves with dots correspond to performance observed with random selection, while the blue ones correspond to the selection made with WSP and finally the green ones to the KS selections. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

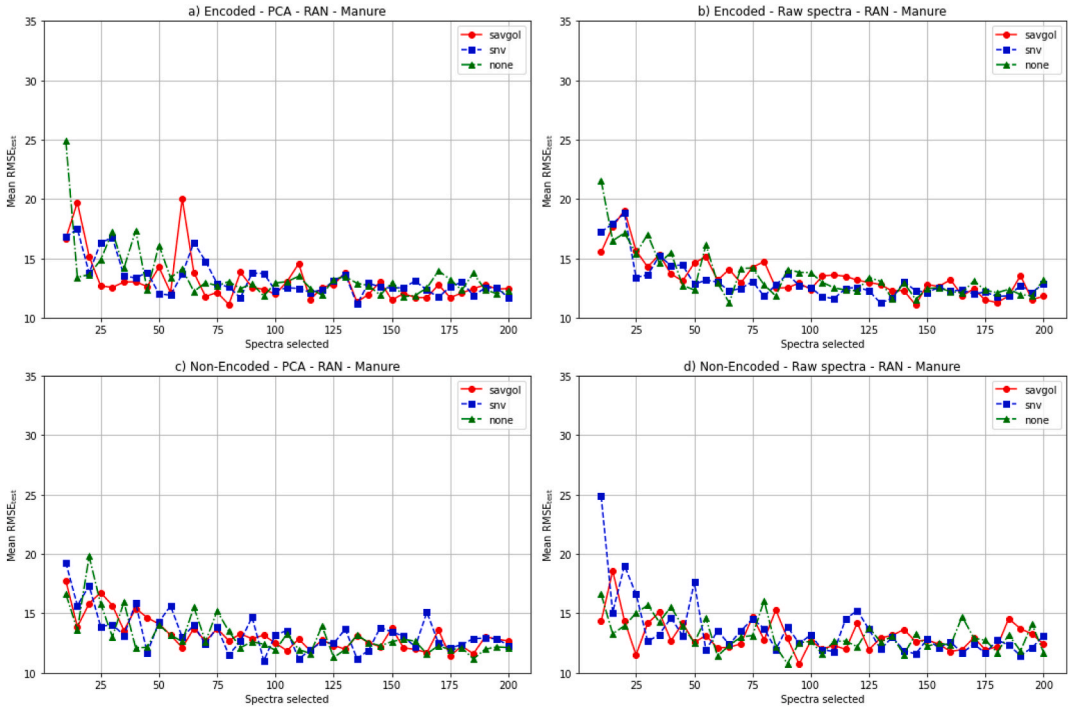


Fig. 7. Plot of the mean RMSE test on the manure dataset for various configurations using WSP a) Configurations C15, C23 and C31; b) Configurations C16, C24 and C32; c) Configurations C14, C22 and C30; d) Configurations C13, C21 and C29. Red curves with dots correspond to the configurations with savgol, while the blue curves with squares correspond to snv and the green ones to no pretreatment. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

3.2.2. *Manure dataset*

Fig. 7a–d shows RMSE of the test dataset for manure dataset, under the configurations C13 to C16, C21 to C24 and C29 to C32 using KS across a range of 40 to 200 selected spectra.

For Fig. 7a, RMSE values for SNV preprocessing range from 16.45 to 12.30, while values for no pretreatment range from 24.94 to 12.43, and Savitzky-Golay preprocessing from 16.43 to 12.60. RMSE decreases notably between 10 and 50 selected spectra, with stabilization observed

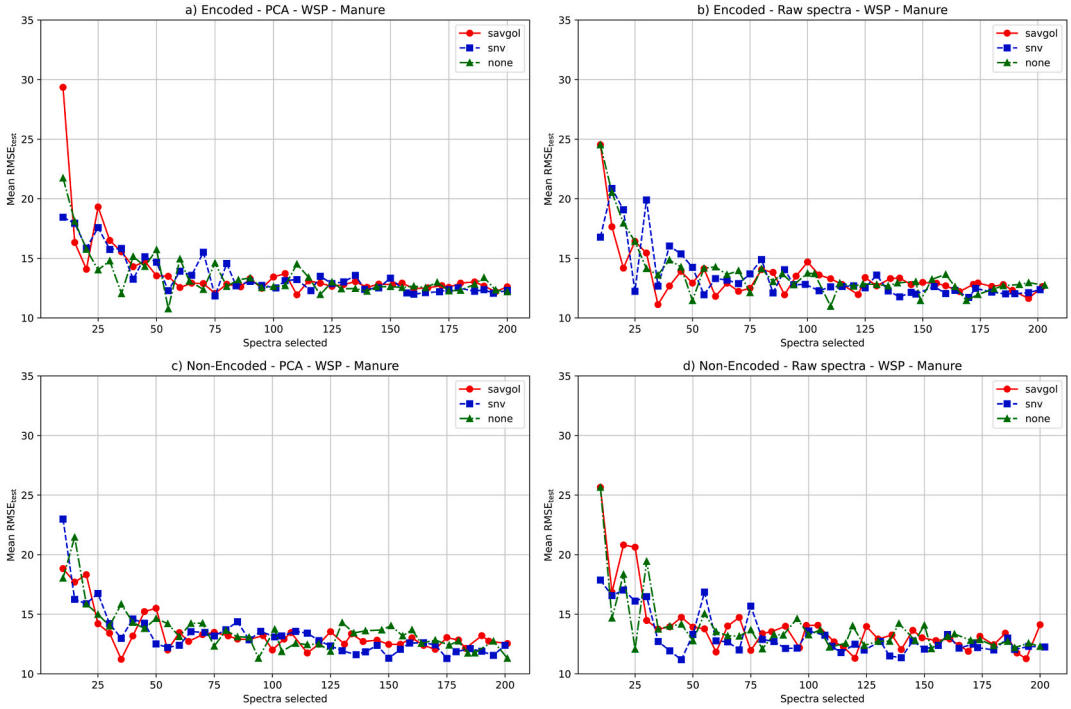


Fig. 8. Plot of the mean RMSE test on the manure dataset for various configurations, over a range of 40 to 200 selected spectra using KS a) Configurations C15, C23 and C31; b) Configurations C16, C24 and C32; c) Configurations C14, C22 and C30; d) Configurations C13, C21 and C29. Red curves with dots correspond to the configurations with savgol, while the blue curves with squares correspond to snv and the green ones to no pretreatment. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

around 100 spectra. No preprocessing method consistently outperforms the others, although modest peak performances are observed for 20-35 spectra with no pretreatment.

Fig. 7b shows RMSE values from 17.13 to 12.92 for SNV, 21.63 to 13.02 for no pretreatment, and 15.24 to 12.32 for Savitzky-Golay. RMSE decreases between 10 and 50 spectra, stabilizing around 100 spectra. Slight peaks are observed for 25, 35, and 40 spectra for the preprocessed methods, suggesting minor benefits of preprocessing when encoding is applied directly to pre-processed spectra rather than PCA scores.

For Fig. 7c, RMSE ranges from 18.93 to 12.33 for SNV, while no pretreatment ranges from 16.81 to 12.16, and Savitzky-Golay from 17.52 to 12.54. RMSE decreases between 10 and 50 selected spectra, with a plateau at 100 spectra. Preprocessing yields modest peak performances for 25-35 spectra, likely due to removal of instrumental noise, while encoding slightly penalizes performance by emphasizing remaining detrimental information.

Fig. 7d shows RMSE values from 24.87 to 12.43 for SNV, 16.25 to 12.22 for no pretreatment, and 14.89 to 12.31 for Savitzky-Golay. RMSE decreases between 10 and 50 spectra, reaching a plateau around 100 spectra. SNV exhibits modest peaks for 35-45 spectra, whereas no pretreatment and Savitzky-Golay show similar trends.

Across Fig. 7a–d, all configurations stabilize around 100 selected spectra. Minor differences at low selection numbers suggest that dimensionality reduction and encoding slightly improve early performance, particularly for SNV. Overall, preprocessing has minimal impact on this dataset, which appears relatively easy to predict. No method consistently outperforms the others, and encoding shows negligible influence. This indicates that, for simpler datasets, WSP achieves sufficient performance with fewer spectra and preprocessing is largely unnecessary, with minimal variation across configurations.

The same analysis is conducted on KS and is shown in Fig. 8a–d.

Fig. 8a–d shows RMSE of the test dataset for manure dataset, under the configurations C13 to C16, C21 to C24 and C29 to C32 using KS across a range of 40 to 200 selected spectra. Contrary to Fig. 7a–d, these figures are displaying more differences between preprocessing. Therefore, it demonstrates that WSP is less sensitive to preprocessing than KS.

For Fig. 8a, RMSE values for SNV preprocessing range from 18.15 to 12.51, while no pretreatment ranges from 21.73 to 12.47, and Savitzky-Golay from 29.63 to 12.63. RMSE decreases notably between 10 and 40 selected spectra, stabilizing around 100 spectra for SNV and no pretreatment. Savitzky-Golay does not reach a plateau in the studied range, outperforming the others from 65 to 200 spectra. This suggests that Savitzky-Golay removes detrimental information more effectively, making PCA-based selection more relevant across the spectrum.

Fig. 8b shows RMSE values from 17.34 to 12.52 for SNV, 24.84 to 12.63 for no pretreatment, and 24.84 to 12.61 for Savitzky-Golay. RMSE decreases between 10 and 50 spectra, stabilizing around 100 spectra for no pretreatment and Savitzky-Golay, while SNV never fully plateaus. The use of raw spectra reduces the differences between preprocessing methods and slightly degrades performance compared to PCA scores.

For Fig. 8c, RMSE ranges from 23.21 to 12.47 for SNV, from 18.21 to 11.35 for no pretreatment, and from 18.76 to 12.50 for Savitzky-Golay. RMSE decreases between 10 and 30 spectra, with stabilization around 125 spectra. Non-encoded PCA scores reduce differences between preprocessing methods but allow slightly better performance for 40-55 spectra.

Fig. 8d shows RMSE values from 17.95 to 12.53 for SNV, 25.91 to 12.49 for no pretreatment, and 25.97 to 14.62 for Savitzky-Golay. RMSE decreases between 10 and 50 spectra, stabilizing around 125 spectra. SNV performs best at low selection numbers in Fig. 8c but lags in 8d, while all methods converge around 125 spectra.

Across Fig. 8a–d, PCA consistently improves early performance for KS by emphasizing dominant linear structures, whereas raw spectra reduce this advantage. Savitzky-Golay is particularly effective when PCA is applied, while variable encoding tends to distort distances, negatively affecting KS. Overall, this dataset requires fewer spectra to reach satisfactory performance than the milk dataset, making differences between preprocessing configurations less pronounced. KS remains more sensitive to preprocessing than WSP, underlining the need to tailor preprocessing choices when using KS.

Fig. 9 shows the mean RMSE on the manure dataset, with the C15 for WSP and C23 for KS over a range of 40 to 200 selected spectra. This figure allows comparison of KS (green curve) and WSP (blue curve) on their best configuration tested with the random selection (red curve) that represents no selection algorithm used. RMSE decreases significantly between 10 and 50 selected spectra until reaching a plateau for 135 selected spectra for all 3 algorithms.

Compared to the milk dataset, the differences between the selection algorithms are less pronounced in this case. This is likely due to the relationship between the spectral data and the target response in this dataset is relatively simple to model. As a result, even random selection is sufficient to capture the relevant information, making it more difficult to clearly distinguish the advantages of KS and WSP.

In conclusion, WSP shows little to no variation in performance across different preprocessing methods for this dataset. This robustness can be attributed not only to WSP's lower sensitivity to preprocessing, but also to the fact that the relationship between the spectral data and the target variable is relatively simple to capture. In such cases, the choice of preprocessing has less impact, especially for algorithms like WSP that aim to evenly explore the spectral space. Nonetheless, WSP exhibits a positive response to the use of PCA and variable encoding, both of which help to slightly enhance or stabilize its performance. On the other hand, KS is more sensitive to preprocessing choices. While PCA has a beneficial effect on KS, likely due to its ability to emphasize dominant patterns in the data, the encoding step has a negative impact, probably because it alters the natural distance relationships that KS depends on to identify extreme samples. These observations reinforce the idea that WSP may be a more robust option in uncertain preprocessing conditions, while KS can benefit more from carefully tuned configurations when the data structure allows it.

4. Conclusion

The study examined the impact of dimensionality reduction, variable encoding, and preprocessing strategies on the performance of sample selection algorithms, notably KS and WSP. Across all datasets and configurations tested, both selection algorithms perform better than the random selection. This highlights the necessity of selecting spectra during model training to achieve good predictive performance at a lower cost. Moreover, the relative gain from selection was proportional to the difficulty of the prediction scenario: the more challenging the dataset, the greater the benefit of using a selection algorithm. As a result, we highly recommend selecting spectra systematically instead of doing reference analysis on whole datasets, particularly for complex applications.

Concerning the parameters influencing the selection, across all simulated datasets, both dimensionality reduction and encoding enhanced the performance of the selection algorithms. PCA and encoding improved the spatial structure which enables a uniform exploration. The combined effect of the two parameters brings the selection algorithms closer to Mahalanobis-distance-based approaches, which help to consider the structure of the data compared to Euclidean-distance-based approaches. Both influences of these parameters on selection were more pronounced in the challenging scenario, which accentuate the need for selection when calibrating a model. However, when applied to the real datasets, it did not follow the same behavior observed on simulated data. On the milk dataset, KS displayed a clear degradation on performance for the configuration where PCA and encoding were applied. One hypothesis is that the PCA changed too much the structure of the data for KS, resulting in bad performance. Finally, this degradation was less pronounced for the manure dataset. As for WSP, it displayed the same behavior as for simulated data for both

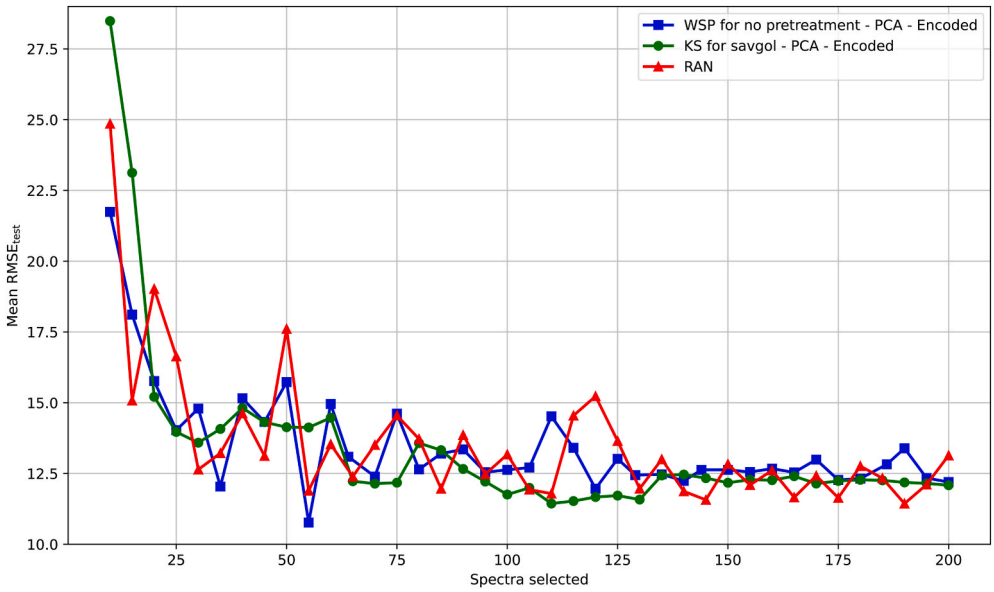


Fig. 9. Plot of RMSE test on the manure dataset for the best configurations using KS, WSP and random selection algorithms; C15 using WSP; b) C23 using KS. Red curves with dots correspond to performance observed with random selection, while the blue ones correspond to the selection made with WSP and finally the green ones to the KS selections. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

datasets. In fact, it was observed that performance fluctuated less with the application of PCA and encoding, which was not observed in the simulated cases.

Regarding preprocessing, the results show that this parameter does indeed impact KS and WSP differently. WSP displayed greater robustness, showing smaller performance variations across preprocessing strategies, while KS was more sensitive and prone to abrupt performance shifts. These behaviors can be attributed to the strategy of both algorithms. WSP by exploring evenly the space is less affected by abrupt changes while KS is directly affected because this algorithm selects extremes samples. Consequently, for users uncertain about the appropriate preprocessing to use in their application, WSP is recommended as the more stable choice. Conversely, if KS is to be used, it is advisable either to apply a carefully selected preprocessing method or to avoid preprocessing altogether.

Further research should focus on the application of sample selection algorithms to nonlinear data structures. Such data generally require larger sample sizes to effectively capture the underlying complexity using more complex models such as neural networks. In this context, sample selection algorithms will become even more valuable, as they can help reduce the number of spectra needed to calibrate such models. A second promising direction involves enhancing selection strategies by moving beyond purely spectral-space-based selection algorithms. A key direction for future research lies in the integration of Active Learning principles [27–29] into spectra selection strategies. Active learning is an iterative loop-based process that goes beyond purely exploratory selection by incorporating criteria such as model uncertainty [29,30]. Each iteration attempts to select new samples based on the knowledge gained from previous iterations and thus consider the need of the current model. Each iteration attempts to select new samples based on the knowledge gained from previous iterations, thereby accounting for what is most needed by the current model. Integrating such principles and criteria like uncertainty with exploratory algorithms like WSP could help strike a balance between diversity and informativeness in spectra selection.

CRediT authorship contribution statement

Carl Charloto: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Maxime Metz:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Nicolas Grotus:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Visualization, Writing – review & editing. **David Esteve:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Michelle Sergent:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Magalie Claeys-Bruno:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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